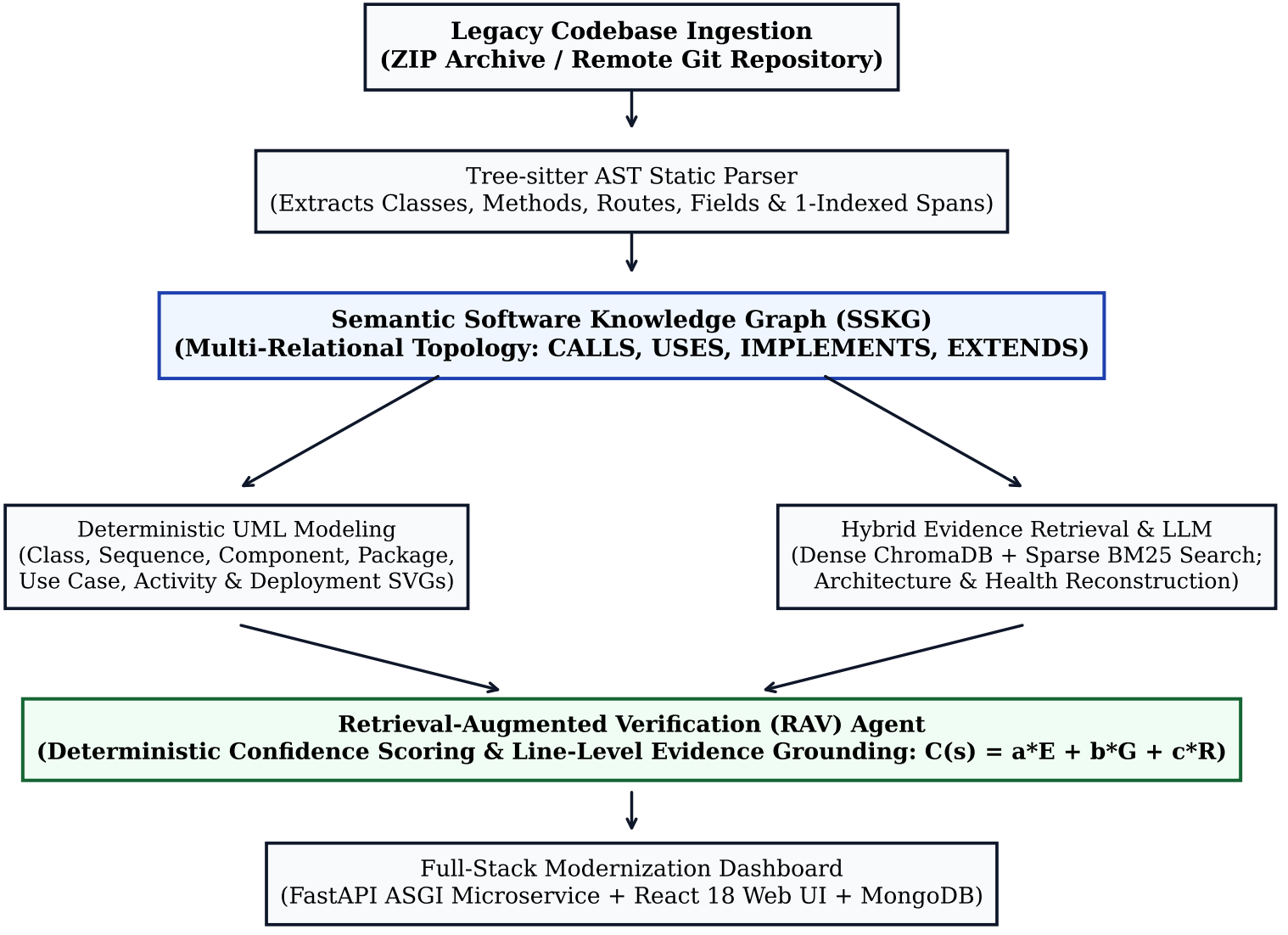
Legacy Codebase Ingestion
(ZIP Archive / Remote Git Repository)
Tree-sitter AST Static Parser
(Extracts Classes, Methods, Routes, Fields & 1-Indexed Spans)
Semantic Software Knowledge Graph (SSKG)
(Multi-Relational Topology: CALLS, USES, IMPLEMENTS, EXTENDS)
Deterministic UML Modeling
(Class, Sequence, Component, Package,
Use Case, Activity & Deployment SVGs)
Hybrid Evidence Retrieval & LLM
(Dense ChromaDB + Sparse BM25 Search;
Architecture & Health Reconstruction)
Retrieval-Augmented Verification (RAV) Agent
(Deterministic Confidence Scoring & Line-Level Evidence Grounding: C(s) = a*E + b*G + c*R)
Full-Stack Modernization Dashboard
(FastAPI ASGI Microservice + React 18 Web UI + MongoDB)